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https://SCOTXWatch.org/ (ISBN 9781971481296)

FORMAL NOTICE OF PROPRIETARY RIGHTS AND ACTUAL KNOWLEDGE OF PENDING CLAIMS UNDER 35 U.S.C. § 154(d) REGARDING VEHICULAR OPTICAL DATA BUSES, V2V OPTICAL COMMUNICATION, AND AUTONOMOUS FLEET LOGISTICS

This Notice: https://storage.googleapis.com/rj-aguiar/notices/AUTO_IP_LIABILITY_NOTICE.pdf

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RE: FORMAL NOTIFICATION OF PENDING PATENT CLAIMS, INITIATION OF PROVISIONAL RIGHTS PERIOD, AND DEMAND FOR DUE DILIGENCE CONCERNING THE AGUIAR MARCANO INTELLECTUAL PROPERTY PORTFOLIO

DATE: April 9, 2026

DEAR: Ms. Barra, Mr. Farley, Mr. Croley, Mr. Filosa, Mr. Fossati, Mr. Musk, Mr. Papin, Mr. Tavi, Mr. Munoz, Mr. Klapper, Mr. Doll, Mr. Guyton, Mr. Murphy, Mr. Yoon, Mr. Close, Mr. Mackensen, Mr. Rawlinson, and Mr. Scaringe,

This comprehensive technical and legal brief is served upon the Chief Executive Officers and Chief Legal Officers of the designated automotive manufacturers to establish formal, legally binding actual notice.

As the leaders of the corporations responsible for the vast majority of automotive assembly and engineering within the United States, your organizations are rapidly transitioning from traditional mechanical manufacturing to the deployment of "Software-Defined Vehicles" (SDVs). Modern automobiles are effectively mobile data centers, processing terabytes of sensor data for Advanced Driver Assistance Systems (ADAS) and autonomous navigation.

However, your industry's roadmap for vehicular networking, Vehicle-to-Vehicle (V2V) communication, and electric vehicle (EV) cybersecurity directly intersects with the proprietary claims of the Aguiar Marcano Intellectual Property Portfolio. This portfolio comprises thirty-three (33) foundational patent applications formally executed and filed with the United States Patent

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and Trademark Office (USPTO) in March 2026.[1]

The primary objective of this document is to satisfy the stringent requirements for "Actual Notice" pursuant to 35 U.S.C. § 154(d), thereby initiating the immediate accrual of provisional rights and the corresponding liability for reasonable royalties.[2]

I. Sector-Specific Liability Mapping: The End of the Copper Wiring Harness

The automotive industry is facing a profound physical crisis: the copper wiring harness is now one of the heaviest and most complex components in a modern vehicle. As you integrate high-resolution LiDAR, radar, and 4K camera arrays to achieve Level 3 and Level 4 autonomy, traditional Controller Area Networks (CAN bus) and even Automotive Ethernet are choking on the bandwidth requirements.

Attempting to scale current electronic architectures will literally make your vehicles too heavy to achieve regulatory range requirements. The inevitable architectural leap your engineering teams must make is legally encumbered by our portfolio.

1. Massively Parallel Optical Data Bus Architecture (USPTO 19/562,818) & The Optical Computing Architecture (USPTO 19/560,038) To solve the vehicular I/O bottleneck, your next-generation platforms will be forced to replace heavy copper wiring harnesses with lightweight, high-capacity optical data fabrics. USPTO 19/562,818 claims an apparatus for an ultra-wide data bus implemented via a plurality of parallel optical channels, effectively turning a microscopic fiber bundle into a multi-terabit data pipe.[1] Furthermore, USPTO 19/560,038 claims the end-to-end integration of this optical fabric to interconnect decentralized processing units and peripheral sensors.[1] Any deployment of parallel optical interconnects to route autonomous driving sensor data to a central compute module directly infringes upon these foundational hardware claims.

II. Vehicle-to-Vehicle (V2V) and Free-Space Telemetry

Your current reliance on 5G and dedicated short-range communications (DSRC) for V2V data sharing is highly vulnerable to RF congestion, spectrum limitations, and latency.

1. High-Bandwidth, Free-Space Optical Data Exchange Between Mobile Devices (USPTO 19/560,042) As vehicles become equipped with high-resolution LED matrix headlights and taillights, the transition to optical V2V communication is inevitable. USPTO 19/560,042 claims a method for ultra-high-speed, free-space data exchange between mobile entities by modulating visual information at high frequencies, utilizing cameras as optical receivers.[1] If your ADAS engineering teams deploy software updates that modulate exterior vehicular lighting to transmit

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telemetry, braking intent, or environmental mapping data to adjacent vehicles via free-space optics, your vehicles become unlicensed optical data routers under this claim.[1]

III. EV Infrastructure Cybersecurity and "Off-Grid" State Enforcement

The proliferation of connected EVs presents an unprecedented national security vulnerability. A compromised charging network could theoretically be used to simultaneously brick millions of vehicles or orchestrate a coordinated drain on the municipal power grid.

1. Hardware-Enforced Grid-Isolated Operation (USPTO 19/574,356) To protect vehicle operational technology (OT) from remote compromise during charging cycles, physical hardware isolation is required. USPTO 19/574,356 claims a security apparatus that physically disconnects network interfaces and high-emission peripherals the exact moment a device draws power from an external electrical grid.[1] Any integration of hardware-level isolation by your EV powertrain divisions—ensuring the vehicle's critical driving systems can only boot or accept firmware updates in a battery-powered, grid-disconnected state—falls squarely under the protection of this patent.[1]

IV. Dual-Purpose Fleet Logistics and Autonomous Robotaxis

Several of your organizations, notably Tesla and General Motors (Cruise), are developing autonomous fleets intended to operate as decentralized ride-hailing networks.

1. Business Method for Dual-Purpose Logistics Networks (USPTO 19/560,069) As connected vehicles generate petabytes of mapping and training data daily, uploading this data over cellular networks is prohibitively expensive and slow. USPTO 19/560,069 claims a method for optimizing a primary logistics network (e.g., passenger transport or autonomous ride-hailing) while simultaneously utilizing those vehicles to physically transport high-capacity data payloads (e.g., Portable Data Cartridges, USPTO 19/560,062) between edge nodes or charging hubs.[1] Utilizing your autonomous fleets to physically "mule" training data back to your central servers while fulfilling passenger routes directly triggers this business method claim.[1]

V. Mandatory Due Diligence Protocols

Under 35 U.S.C. § 154(d), the statutory clock for the calculation of retroactive reasonable royalties commences immediately upon your receipt and actual knowledge of the published claims detailed herein.[2]

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The executive leadership of the addressed automotive manufacturers is hereby directed to execute the following immediate remediation protocols:

1. **Initiation of FTO Audits:** The respective General Counsels and Chief Legal Officers must immediately retain specialized external patent counsel to cross-reference the thirty-three (33) applications within the Aguiar Marcano Portfolio against all current R&D roadmaps for optical wiring harnesses, EV charging cybersecurity protocols, and V2V optical telemetry.
2. **Quarantine of Intersecting Architectures:** Any planned deployment of multi-channel optical data buses to replace copper ADAS sensor wiring, or the use of LED matrix lighting for free-space data transmission, must be suspended or quarantined pending legal clearance.

VI. Strategic Licensing and Ultimatum

The technological methodologies claimed in the Aguiar Marcano Portfolio are the fundamental prerequisites for the next era of automotive engineering. Attempting to overcome the physical weight of copper wiring or the latency of RF communication without intersecting these patents is mathematically and physically improbable.

We formally invite your respective internal legal departments to initiate a technical review of the cited USPTO applications. A formal response outlining your intended compliance audit, a presentation of non-infringement theories, or a request to discuss an industry-wide licensing framework is expected within thirty (30) business days of the transmission of this notice.

Govern yourselves accordingly.

Reinaldo Aguiar Marcano

Works cited

1. 04-ISBN-978-1-97-148129-6__Portfolio-of-Patent-Applications-filed-with-the-USPTO.pdf
2. 35 USC 154 Contents and term of patent; provisional rights. - BitLaw, accessed April 9, 2026, https://www.bitlaw.com/source/35usc/154.html
3. Preserving Provisional Rights for Pre-Issuance Patent Damages, accessed April 9, 2026, https://www.hsfkramer.com/kl-pdfs/4/3/43719.pdf

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